SECTION B

Option A – Alternating Currents

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)	Reactance (accept impedance/resistance) of inductor increases with frequency (1) Resistor takes a smaller portion of pd (**OR** current decreases so pd across R decreases) (1)	2			2		
		(ii)	Equating $\omega L = R$ (1) Answer = 141 k[Hz] (1)		2		2	2	
		(iii)	Correct method (implied by correct answer) (1) Answer = 1.77 [V] (1.25 [V] no marks) (1)		2		2	1	
	(b)	(i)	Flux proportional to B (1) Flux proportional to A (1) Same flux for each turn (N) (1) **OR** using Faraday's e.g. $V = \frac{\mathrm{d}}{\mathrm{dt}}(BAN\cos\theta)$ (1) Taking B and A outside bracket (1) Stating B and A are constants (1) **OR** Faraday's law stated (1) Correct explanation of rate of change of flux with reference to cutting of field lines, linked to B (1) Correct explanation of rate of change of flux with reference to cutting of field lines, linked to A (1)	3			3		
		(ii)	When flat (or wtte) because cutting at right angles **OR** flat because max rate of change of flux	1			1		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	Answer = 152 m[A]		1		1	1	
		(ii)	Correct angular frequency obtained (28 072) **OR** f_0 = 4468 [Hz] **OR** 2 × f_0 = 8932 [Hz] (1) Correct use of impedance equation i.e.1140 [Ω] (1) Rearrangement of $I = \frac{V}{Z}$ (1) **ecf** on Z Correct answer = 11 m[A] (1)		4		4	4	
	(d)		Multiplying 240 [mV] by $\sqrt{2}$ [= 339 [mV]] (1) Dividing by y-sensitivity (**ecf** on 240) (1) Using $T = \frac{1}{f}$ (14.8 μs) (1) Dividing by time div (7.4 squares) (1) Time base good (one and 1/3 wave ok) **AND** y-sensitivity gives too small a height (independent mark & may well imply the other 4 marks if seen alone) (1)			5	5	2	
			Question 7 total	**6**	**9**	**5**	**20**	**10**	**0**